

The HIV Treatment Gap: Estimates of the Financial Resources Needed versus Available for Scale-Up of Antiretroviral Therapy in 97 Countries from 2015 to 2020

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Evidence abstract

BACKGROUND The World Health Organization (WHO) released revised guidelines in 2015 recommending that all people living with HIV, regardless of CD4 count, initiate antiretroviral therapy (ART) upon diagnosis.

However, few studies have projected the global resources needed for rapid scale-up of ART.

Under the Health Policy Project, we conducted modeling analyses for 97 countries to estimate eligibility for and numbers on ART from 2015 to 2020, along with the facility-level financial resources required.

We compared the estimated financial requirements to estimated funding available.

METHODS AND FINDINGS Current coverage levels and future need for treatment were based on country-specific epidemiological and demographic data.

Simulated annual numbers of individuals on treatment were derived from three scenarios: (1) continuation of countries' current policies of eligibility for ART, (2) universal adoption of aspects of the WHO 2013 eligibility guidelines, and (3) expanded eligibility as per the WHO 2015 guidelines and meeting the Joint United Nations Programme on HIV/AIDS "90-90-90" ART targets.

We modeled uncertainty in the annual resource requirements for antiretroviral drugs, laboratory tests, and facility-level personnel and overhead.

We estimate that 25.7 (95% CI 25.5, 26.0) million adults and 1.57 (95% CI 1.55, 1.60) million children could receive ART by 2020 if countries maintain current eligibility plans and increase coverage based on historical rates, which may be ambitious.

If countries uniformly adopt aspects of the WHO 2013 guidelines, 26.5 (95% CI 26.0 27.0) million adults and 1.53 (95% CI 1.52, 1.55) million children could be on ART by 2020.

Under the 90-90-90 scenario, 30.4 (95% CI 30.1, 30.7) million adults and 1.68 (95% CI 1.63, 1.73) million children could receive treatment by 2020.

The facility-level financial resources needed for scaling up ART in these countries from 2015 to 2020 are estimated to be US\$45.8 (95% CI 45.4, 46.2) billion under the current scenario, US\$48.7 (95% CI 47.8, 49.6) billion under the WHO 2013 scenario, and US\$52.5 (95% CI 51.4, 53.6) billion under the 90-90-90 scenario.

After projecting recent external and domestic funding trends, the estimated 6-y financing gap ranges from US\$19.8 billion to US\$25.0 billion, depending on the costing scenario and the U.S.

President's Emergency Plan for AIDS Relief contribution level, with the gap for ART commodities alone ranging from US\$14.0 to US\$16.8 billion.

The study is limited by excluding above-facility and other costs essential to ART service delivery and by the availability and quality of country- and region-specific data.

CONCLUSIONS The projected number of people receiving ART across three scenarios suggests that countries are unlikely to meet the 90-90-90 treatment target (81% of people living with HIV on ART by 2020) unless they adopt a test-and-offer approach and increase ART coverage.

Our results suggest that future resource needs for ART scale-up are smaller than stated elsewhere but still significantly threaten the sustainability of the global HIV response without additional resource mobilization from domestic or innovative financing sources or efficiency gains.

As the world moves towards adopting the WHO 2015 guidelines, advances in technology, including the introduction of lower-cost, highly effective antiretroviral regimens, whose value are assessed here, may prove to be "game changers" that allow more people to be on ART with the resources available.

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Evidence checklist

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2. However, few studies have projected the global resources needed for rapid scale-up of ART.
3. Under the Health Policy Project, we conducted modeling analyses for 97 countries to estimate eligibility for and numbers on ART from 2015 to 2020, along with the facility-level financial resources required.
4. We compared the estimated financial requirements to estimated funding available.
5. **METHODS AND FINDINGS** Current coverage levels and future need for treatment were based on country-specific epidemiological and demographic data.
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7. We modeled uncertainty in the annual resource requirements for antiretroviral drugs, laboratory tests, and facility-level personnel and overhead.
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9. If countries uniformly adopt aspects of the WHO 2013 guidelines, 26.5 (95% CI 26.0 27.0) million adults and 1.53 (95% CI 1.52, 1.55) million children could be on ART by 2020.
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11. The facility-level financial resources needed for scaling up ART in these countries from 2015 to 2020 are estimated to be US\$45.8 (95% CI 45.4, 46.2) billion under the current scenario, US\$48.7 (95% CI 47.8, 49.6) billion under the WHO 2013 scenario, and US\$52.5 (95% CI 51.4, 53.6) billion under the 90-90-90 scenario.
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